Int J Adv Manuf Technol (2013) 65:183–196
DOI 10.1007/s00170-012-4159-3

ORIGINAL ARTICLE

Multi-objective stochastic programming to solve manpower allocation problem

Mostafa Ekhtiari · Keivan Ghoseiri

Received: 15 June 2010 / Accepted: 10 April 2012 / Published online: 5 May 2012
© Springer-Verlag London Limited 2012

Abstract Effective manpower allocation is among the most vital and complicated decisions for most companies on account of imprecise nature of input information of the problem. This paper presents a novel combination of the chance-constrained programming and the global criterion model for manpower allocation problem that is called chance-constrained global criterion. The proposed model is a deterministic equivalent for the multi-objective stochastic problem of manpower allocation. To illustrate the model, a tri-objective stochastic manpower allocation case problem for determining optimal number of manpower in a job-shop manufacturing system is formulated and solved, and then the competitive advantages of the model are discussed. To have a better judgment on the validity and performance efficiency of the model, 20 different problems are generated and solved. The results show that increasing the size of problem do not have much effect on the number of iterations required for finding the optimal solution, and this model decreases complicacy in modeling the problem.

Keywords Chance-constrained programming approach · Global criterion · Multi-objective stochastic programming · Manpower allocation problem

M. Ekhtiari (✉)
Faculty of Industrial and Mechnical Engineering, Qazvin Branch,
Islamic Azad University,
Qazvin, Iran
e-mail: m_ektiari@yahoo.com

K. Ghoseiri
Department of Civil and Environmental Engineering,
University of Maryland,
College Park, MD 20742-3021, USA
e-mail: kghoseir@umd.edu

1 Introduction

Manpower as one of the most important organizational resources plays a key role in increasing the efficiency of organizations and can realize the strategic goals of them. Therefore, effective manpower allocation planning is very important. On the other hand, future enlargement of the organizations in competitive environments must be considered.

Over the last 25 years, there has been substantial growth in the development of mathematical models for use in manpower planning. A number of these models have included optimization features, generally involving either linear programming (LP) or network optimization. Since the allocation process involves the evaluation of candidates (i.e., qualified employees), based on a variety of criteria, it is a typical multi-criteria decision making (MCDM) problem [1]. Several criteria, such as problem-solving skill, technical skill, human relationship skill, previous work achievement, and personality characteristics, have to be taken into account in the assignment process. For example, Caballero et al. [2] proposed a methodology, which admits the incorporation of efficiency and quality criteria in the planning and allocation of new human academic resources within a university, based on the combination of two quantitative techniques, which provide support for decision making: data envelopment analysis and MCDM.

Kwak and Lee [3] presented a specific mathematical programming model, which can be used to perform the effective resource allocation of limited human resources in a health care system. Summers [4] used a linear programming approach to solve the audit staff-assigning problem. He saw the limitation of linear programming, which focused on benefit maximization and assumed that the audit office was trying to maximize a mixture of monetary and non-monetary benefits. Therefore, the key factor for his model was assigning monetary equivalent benefits for several non-monetary benefits.

Killough and Souders [5] used a goal programming approach to incorporate multiple objectives. Bailey et al. [6] advocated the use of the goal programming approach for the audit staff allocation problem against a linear programming approach. They solved Summers' model using goal programming, and the results showed that goal programming provided better solutions by incorporating several goals compared with solutions of linear programming. The goal programming formulation permits explicit consideration of trade-offs in changing priority of goals. Welling [7] also used a goal programming approach to take into account human resource interactions in an audit staff allocation problem. He used job productivity, human resource development, and individual satisfaction as non-monetary benefits in human resource valuation. Balachandran and Steuer [8] used a multiple objective linear programming (MOLP) model for the audit staff allocation problem. They discussed that goal programming had limitations because weights should be assigned a priori. However, in the GP, there is no clear relation between the weights and the solution found as you might end up in new vertices in both of the solution space and utopian set [9]. In addition, goal programming cannot deal with the organizational differentiation problems [10].

Gardner et al. [11] also proposed a multi-period MOLP approach to solve audit staff planning problems. Kwak et al. [10] proposed a model of tri-objective with some constraints to allocate human resource in a Certified Public Accountants for a single period model. Chebat et al. [12] proposed an approach to evaluate and audit the efficiency of human and financial resources in marketing. Pan et al. [13] presented an approach using mixed-integer programming with a two-stage heuristic algorithm for solving the manpower scheduling problem of the high mix low-volume job shop of the precision engineering industry.

However, in many situations, it is very difficult to describe these criteria by means of clear-cut values because they are subjective, vague, and based on imprecise information [1]. For example, Liang and Wang [14] used the fuzzy set theory introduced by Zadeh [15] and the weighted complete bipartite graph to solve the personnel placement problem. Toroslu [16] proposed a variation of the personnel allocation problem with some ordering constraints on the partitions of the bipartite graph. Liang and Wang [17] also developed a fuzzy MCDM algorithm that used both subjective and objective assessments to select employees.

Yaakob and Kawata [18] considered the number of workers required by each job and the relationships among all workers. Karsak [19] proposed an approach of fuzzy multiple objective programming for selecting employees. They allocated suitable workers to each job in an industrial environment. As the aforementioned literature revealed, most of the existing personnel assignment approaches primarily optimize the single criterion of weighted composite scores of personnel performance. Huang et al. [1] was one of the few that proposed a bi-objective model for assigning candidates to appropriate positions. They proposed a binary integer programming model to maximize the total weighted composite scores and simultaneously minimize the total difference between the scores. The model was transformed into a fuzzy goal programming, which could be used to solve personnel assignment problem.

As the literature review shows, this research is the first of its kind that proposes stochastic programming as the most popular approach under uncertainty conditions to model and solution of the manpower allocation problem. Stochastic programming deals with a class of optimization models and algorithms, in which all or some of the parameters may be subject to significant uncertainty [20]. The models of stochastic programming yield plans that are better able to hedge against losses and catastrophic failures [21]. Because of these properties, stochastic programming models have been developed for a variety of applications, including electric power generation [22], financial planning [23], telecommunications network planning [24], flexible manufacturing systems [25], supply chain management [26–28], and aggregate production planning [29]. The widespread applicability of stochastic programming models has attracted considerable attention from the OR/MS community resulting in several recent books [30–32] and survey articles [33, 34]. The most popular approach to solve the multi-objective stochastic problem is the chance-constrained programming (CCP) [35, 36] approach. CCP approach transforms multi-objective stochastic problem into a multi-objective deterministic equivalent problem, which can easily be solved by an appropriate optimization technique. CCP attempts to maximize the expected value of random objectives while a given confidence level of satisfaction for random constraints is secured.

Industrial and manufacturing environments are subject to uncertain events, which may be caused by both human and machine resource factors such as worker unavailability, failure to take delivery of raw material, change in availability dates, machine failure, and change in the customer orders. Hence, considering a scheduling problem in a stochastic situation is more realistic than in a deterministic one [37]. The job-shop scheduling is one of the most typical and complicated scheduling problems. Some of the latest works concerning stochastic job-shop scheduling problems are Lawrence and Sewell [38], Singer [39], Yoshitomi [40], Yoshitomi and Yamaguchi [41], Tavakkoli-Moghaddam et al. [42], Luh et al. [43], Neumann and Schneider [44], Alcaide et al. [45], and Ahmadizar et al. [37] that are all single objective models.

The global criterion (GC) and compromise programming (CP) are two well-known existing models in "no-preference" category that can be used to solve multi-objective problems when the necessary conditions met. One of their advantages is that they do not need any DM available for decision making in the multi-objective problems. In both GC and CP models, the objective is minimizing the deviations from the ideal solution. For measuring the distance between Pareto optimal point and

ideal point in the GC and CP, different metrics are proposed. In these metrics, using higher value for metric l_P increases the effectiveness of GC and CP methods in providing the complete Pareto optimal set [46].

By considering the advantages of metric $l_{P=\infty}$, the GC compared to the CP has fewer variables and constraints, so that the difference between the number of variables and constraints of these models is equal to objectives' number of the main multi-objective problem. Abdelaziz et al. [20] proposed a deterministic equivalent model for solving the multi-objective stochastic problems, which is called as chance-constrained compromise programming (CCCP). The CCCP model is formulated by combining the CCP approach and the model of CP.

In this paper, we propose a deterministic equivalent of multi-objective stochastic problem that combines the well-known approaches of CCP and GC. The proposed model is called chance-constrained global criterion (CCGC) model. One of the advantages of the CCGC model is decrease in the number of variables and constraints of the equivalent deterministic model. Compared to CCCP, decrease in the number of variables and constraints of CCGC model compared to CCCP equals to the number of objectives of main multi-objective problem, and this advantage is of particular importance for modeling of real multi-objective stochastic problems. This paper continues as below.

Section 2 presents nomenclature and definitions. Section 3 introduces the main structure of multi-objective stochastic problem for manpower allocation problem and the CCP approach to solve it. The models of CP and GC are reviewed in Section 4. Section 5 formulates the proposed model of CCGC. In Section 6, the CCGC (min–max) model is illustrated by a multi-objective stochastic problem for manpower allocation in a job-shop manufacturing system. The concluding remarks are presented in Section 7.

2 Nomenclature

$f_k(\mathbf{x})$	kth maximization objective function, k=1,…,K
$f_r(\mathbf{x})$	rth minimization objective function, r=K+1,…,R
x_j	Decision variable for the number of manpower employed in work unit j, j=1,…,n
$\mathbf{x}=(x_1,\ldots,x_n)$	Vector of decision variables, which is part of the solution space (S) of the problem
X	Total of employed manpower in all work units
c_{kj}, c_{rj}, a_{ij}, b_i	Certainty parameters of manpower allocation problem
$\widetilde{c}_{kj}$, $\widetilde{c}_{rj}$, $\widetilde{a}_{ij}$, $\widetilde{b}_i$	Random parameters of manpower allocation stochastic problem
α_i	The maximum allowable probability that constraint i, i=1,…,m
f_k^*	Utopia value associated with maximization objective k, k=1,…,K
$g_i(\mathbf{x})$	ith system constraint function, i=1,…,m
w_k	Preference weight of objective k, k=1,…,K
δ_k^-	Deviational variable of slack for constraint related to kth objective, k=1,…,K
P	Parameter of final utility function
c_{kj}^*	The maximum value observed for objective kth, variable jth for all the state of nature ω
α_k	The threshold value of objective kth, k=1,…,K
w_i	Preference weight of constraint i, i=1,…,m
$\widetilde{o}_j$	Random variable of output quantity per worker in jth work unit, j=1,…,n
w_j	Daily wage/worker employed in work unit jth, j=1,…,n
$\widetilde{t}_j$	Random variable of unallowable idle time/worker employed in jth work unit, j=1,…,n
$x_{j(\text{existent})}$	Number of existent manpower in jth work unit, j=1,…,n

3 The multi-objective stochastic of manpower allocation problem

A general model for the manpower allocation multi-objective problem under unlikely conditions can be stated as follows:

$$\begin{aligned}
&\max f_k(\mathbf{x}) = \textstyle\sum_{j=1}^{n} c_{kj}x_j,\ k = 1,\ldots,\ K,\\
&\min\ f_r(\mathbf{x}) = \textstyle\sum_{j=1}^{n} c_{rj}x_j,\ r = K+1,\ldots,\ R,\\
&\text{subject to}\\
&\sum_{j=1}^{n} a_{ij}x_j \le b_i,\ i = 1,\ldots,m,\\
&\sum_{j=1}^{n} x_j = X,\\
&\mathbf{x} \in S.
\end{aligned} \tag{1}$$

The objectives of program 1 can be generally divided into two categories: Some objectives are positive, and the system aims to maximize them. Some of these objectives are profit, efficiency, and/or general output of the system, so that the aim is to allocate power to the units, which increase the profitability and efficiency of the system. The other category of the objectives is of negative type with the aim of minimizing. Some of these objectives are unauthorized idle time and wage cost. Therefore, our objective is to allocate manpower to the units that decrease unauthorized time and cost

of the system. The constraints of the model expresses minimum or maximum ceiling of the existing resources, upon which the objectives of the problem are satisfied. Some of these constraints include the maximum budget being considered for all units of the system and/or the constraint that can consider maximum existing space for absorbing manpower. Meanwhile, some other constraints can be defined that can consider the manpower required for the whole system and each and every one of working units. Now, considering the collection of contradictorily positive and negative objectives and systemic constraints, the objective is to allocate manpower to all units in such a way that, in addition to provision of constraints, the objectives are appropriately optimized.

In program 1, $\sum_{j=1}^{n} a_{ij}x_j \le b_i,\ i=1,\ldots,m$ are system constraints, and $\sum_{j=1}^{n} x_j = X$ is allocation constraint. Now we assume the parameters of program 1 is not defined certainly and $\widetilde{c}_{kj}$, $\widetilde{a}_{ij}$, and $\widetilde{b}i$ be random parameters. If we maximize the objectives $\sum_{j=1}^{n} \widetilde{c}_{kj}x_j$ (for k=1,…,K) and minimize $\sum_{j=1}^{n} \widetilde{c}_{rj}x_j$ (for r=K+1,…,R), the multi-objective stochastic of manpower allocation problem can be written as follows:

$$\begin{aligned}
&\max\ \textstyle\sum_{j=1}^{n} \widetilde{c}_{kj}x_j,\ k=1,\ldots,\ K,\\
&\min\ \textstyle\sum_{j=1}^{n} \widetilde{c}_{rj}x_j,\ r=K+1,\ldots,\ R,\\
&\text{subject to}\\
&\sum_{j=1}^{n} \widetilde{a}_{ij}x_j \le \widetilde{b}_i,\ i=1,\ldots,m,\\
&\sum_{j=1}^{n} x_j = X,\\
&\mathbf{x}\in S,
\end{aligned} \tag{2}$$

where program 2 is the likely form of the problem introduced in program 1.

In the CCP approach, the program 2 by considering only maximizing objectives is converted into a deterministic equivalent program as follows [32]:

$$\begin{aligned}
&\max\ E\left(\textstyle\sum_{j=1}^{n} \widetilde{c}_{kj}x_j\right),\ k=1,\ldots,\ K,\\
&\text{subject to}\\
&\text{Prob}\left(\sum_{j=1}^{n} \widetilde{a}_{ij}x_j \le \widetilde{b}_i\right) \ge 1-\alpha_i,\ i=1,\ldots,\ m,\\
&\sum_{j=1}^{n} x_j = X,\\
&\mathbf{x}\in S,
\end{aligned} \tag{3}$$

where $E\left(\sum_{j=1}^{n} \widetilde{c}_{kj}x_j\right)$ is the expected value of the objectives with regard to random coefficients $\widetilde{c}_{kj}$. Let α_i (for i=1,…, m) be the maximum allowable probability that constraint i is violated, a chance-constrained programming formulation of LP, say CCP, can be obtained with the chance constraints: $\text{Prob}\left(\sum_{j=1}^{n} \widetilde{a}_{ij}x_j \le \widetilde{b}_i\right) \ge 1-\alpha_i,\ i=1,\ldots,\ m.$

Program 3 can be solved by an appropriate optimization technique. The next section introduces the CP model and then proposes a chance-constrained global criterion model to solve multi-objective stochastic problems.

4 The CP model

The CP model was proposed by Zeleny [47, 48] and Yu [49]. It consists of minimizing the distance between the achievement levels $f_k(\mathbf{x})$ and the utopia values f_k^* associated with each objective k. In the case that the more of the objective is better, the utopia values f_k^* can be obtained as follows:

$$\begin{aligned}
&f_k^* = \max f_k(\mathbf{x}),\ k=1,\ldots,\ K,\\
&\text{subject to}\\
&g_i(\mathbf{x}) \le b_i,\ i=1,\ldots,\ m,\\
&\mathbf{x}\in S.
\end{aligned} \tag{4}$$

Utopia values are the best and most ideal results obtained from each objective function in the constraints of the problem solution space.

The final model of CP by considering preference weights of objectives (w_k) can be formulated as follows:

$$\begin{aligned}
&\min \left\{\textstyle\sum_{k=1}^{K} w_k\left(\delta_k^-\right)^P\right\}^{\frac{1}{P}},\\
&\text{subject to}\\
&f_k(\mathbf{x}) + \delta_k^- = f_k^*,\ k=1,\ldots,\ K,\\
&g_i(\mathbf{x}) \le b_i,\ i=1,\ldots,\ m,\\
&\mathbf{x}\in S,\\
&\delta_k^- \ge 0,\ k=1,\ldots,\ K,
\end{aligned} \tag{5}$$

where the variable δ_k^- is deviational variable of slack for constraint related to kth objective.

The CP model is based on the choice of the solutions that are closer to the ideal points [50, 51]. In the CP model, P is parameter of final utility function, which can have values of metrics $\{1,2,\ldots\}\cup\{\infty\}$. Also w_k>0(for k=1,…,K) and $\sum_{k=1}^{K} w_k = 1$.

The $l_{p=1}$, $l_{p=2}$, and $l_{P=\infty}$ are named Manhattan, Euclidean, and Tchebycheff metrics, respectively. By considering $l_{P=\infty}$ metric, the CP (min–max) model related to program 5 can be formulated as follows:

$$\begin{aligned}
&\min y,\\
&\text{subject to}\\
&(1): y \ge \sum_{k=1}^{K} w_k\delta_k^-,\\
&(2): f_k(\mathbf{x}) + \delta_k^- = f_k^*,\ k=1,\ldots,\ K,\\
&g_i(\mathbf{x}) \le b_i,\ i=1,\ldots,\ m,\\
&\mathbf{x}\in S,\\
&y \ge 0; \delta_k^- \ge 0,\ k=1,\ldots,\ K,
\end{aligned} \tag{6}$$

where y is always non-negative.

Constraint 2 of program 6 can be stated as follows:

$$\delta_k^- = f_k^* - f_k(\mathbf{x}),\ k = 1, \ldots, K, \tag{7}$$

where if Eq. 7 is replaced with constraint 1 of program 6, then we have:

$$y \geq \sum_{k=1}^{K} w_k \left(f_k^* - f_k(\mathbf{x})\right),\ k = 1, \ldots, K. \tag{8}$$

In fact Eq. 8 includes Eq. 7, too. Because program 6 has been formulated with assumption $P = \infty$, we can eliminate constraints 1 and 2 of program 6 and can replace only Eq. 8 with these constraints. The program obtained, which is the well-known model of global criterion with $P = \infty$ (GC (min–max)) and under the assumption of preference weights of objectives (w_k), is stated as follows [52–55]:

$$\begin{aligned} &\min y, \\ &\text{subject to} \\ &y \geq \sum_{k=1}^{K} w_k \left(f_k^* - f_k(\mathbf{x})\right), \\ &\mathbf{x} \in S, \\ &y \geq 0. \end{aligned} \tag{9}$$

The differences of programs 6 and 9 can be stated as advantages of the GC (min–max) model compared to the CP (min–max) model. These advantages are as follows:

1. The fewer the number of variables; considering elimination of variables δ_k^-, program 9 has less k variables.
2. The fewer number of constraints; program 9 by considering elimination of constraints $f_k(\mathbf{x}) + \delta_k^- = f_k^*$ has less k constraint.

Both these advantages can lead to decreased solution time of problems such as large-scale problems and to increased modeling accuracy of these problems, so that the advantages will help planners make better and easier decisions.

5 Chance-constrained global criterion (min–max)

Abdelaziz et al. [20] proposed a model based on combining the CCP approach and the CP model. They introduced their deterministic equivalent programming model for solving multi-objective stochastic problems, so that obtained approach is called as CCCP.

In this section based on the cited advantages about the GC (min–max) model, we propose the CCGC (min–max) model and present how to model the proposed model elaborately. In the following, we show how to transform the random constraints and the random objectives.

5.1 Random constraints and objectives

Random constraints are handled as in the CCP approach: $\widetilde{a}_{ij}$ and $\widetilde{b}_i$ are random and normally distributed parameters. The related chance constraint to $\sum_{j=1}^{n} \widetilde{a}_{ij} x_j \leq \widetilde{b}_i,\ i = 1, \ldots, m$, is:

$$\text{Prob}\left(\sum_{j=1}^{n} \widetilde{a}_{ij} x_j \leq \widetilde{b}_i\right) \geq 1 - \alpha_i,\ i = 1, \ldots, m. \tag{10}$$

Let $\widetilde{A}(\mathbf{x}) = \sum_{j=1}^{n} \widetilde{a}_{ij} x_j - \widetilde{b}_i$, $\widetilde{A}_i(\mathbf{x})$ is normally distributed, and let $E\left(\widetilde{A}_i(\mathbf{x})\right)$ and $\text{Var}\left(\widetilde{A}_i(\mathbf{x})\right)$ be, respectively, the mean and the variance. Thus we have:

$$\text{Prob}\left(\widetilde{A}_i(\mathbf{x}) \leq 0\right) \geq 1 - \alpha_i,\ i = 1, \ldots, m, \tag{11}$$

and then

$$\text{Prob}\left(\frac{\widetilde{A}_i(\mathbf{x}) - E\left(\widetilde{A}_i(\mathbf{x})\right)}{\sqrt{\text{Var}\left(\widetilde{A}_i(\mathbf{x})\right)}} \leq \frac{-E\left(\widetilde{A}_i(\mathbf{x})\right)}{\sqrt{\text{Var}\left(\widetilde{A}_i(\mathbf{x})\right)}}\right) \geq 1 - \alpha_i,\ i = 1, \ldots, m, \tag{12}$$

where $\Phi(z) = P(N(0,1) \leq z)$ represents the cumulative distribution function of a standard normal distribution.

$$\frac{-E\left(\widetilde{A}_i(\mathbf{x})\right)}{\sqrt{\text{Var}\left(\widetilde{A}_i(\mathbf{x})\right)}} \geq \Phi^{-1}(1 - \alpha_i),\ i = 1, \ldots, m, \tag{13}$$

$$E\left(\widetilde{A}_i(\mathbf{x})\right) + \Phi^{-1}(1 - \alpha_i)\sqrt{\text{Var}\left(\widetilde{A}_i(\mathbf{x})\right)} \leq 0,\ i = 1, \ldots, m. \tag{14}$$

and we have:

$$E\left(\sum_{j=1}^{n} \widetilde{a}_{ij} x_j - \widetilde{b}_i\right) + \Phi^{-1}(1 - \alpha_i)\sqrt{\text{Var}\left(\sum_{j=1}^{n} \widetilde{a}_{ij} x_j - \widetilde{b}_i\right)} \leq 0,\ i = 1, \ldots, m. \tag{15}$$

Now, if the objective be minimizing y ($y \geq 0$) subject to:

$$E\left(\sum_{j=1}^{n} \widetilde{a}_{ij} x_j - \widetilde{b}_i\right) + \Phi^{-1}(1 - \alpha_i)\sqrt{\text{Var}\left(\sum_{j=1}^{n} \widetilde{a}_{ij} x_j - \widetilde{b}_i\right)} + y \geq 0,\ i = 1, \ldots, m. \tag{16}$$

then finally it can be said:

$$y \geq E\left(\widetilde{b}_i - \sum_{j=1}^{n} \widetilde{a}_{ij} x_j\right) - \Phi^{-1}(1 - \alpha_i)\sqrt{\text{Var}\left(\sum_{j=1}^{n} \widetilde{a}_{ij} x_j - \widetilde{b}_i\right)},\quad i = 1, \ldots, m. \tag{17}$$

$\widetilde{c}_{kj}$ are random and normally distributed parameters. Let us consider c_{kj}^* as the maximum value observed for objective

kth, variable jth for all the state of nature $\omega\left(c_{kj}^{*}=\max_{\omega}\widetilde{c}_{kj}\right)$. f_k^* is the best solution of the objective function $\sum_{j=1}^{n} c_{kj}^{*}x_j$ subject to system constraints. The kth objective is supposed to be maximized:

$$\begin{aligned} &f_k^* = \max \sum_{j=1}^{n} c_{kj}^{*}x_j,\ k=1,\ldots,K,\\ &\text{subject to}\\ &\mathbf{x}\in S. \end{aligned} \tag{18}$$

By considering program 18, it can be said:

$$\sum_{j=1}^{n}\widetilde{c}_{kj}x_j \le f_k^*,\ k=1,\ldots,K. \tag{19}$$

Based on GC (min–max) method, the objective is minimizing y ($y\ge 0$) subject to:

$$\text{Prob}\left(\sum_{j=1}^{n}\widetilde{c}_{kj}x_j + y \ge f_k^*\right) \ge 1-\alpha_k,\ k=1,\ldots,K, \tag{20}$$

where α_k is the threshold value of objective kth.

$$\text{Prob}\left(f_k^* - \sum_{j=1}^{n}\widetilde{c}_{kj}x_j \le y\right) \ge 1-\alpha_k,\ k=1,\ldots,K, \tag{21}$$

Let $\widetilde{B}_k(\mathbf{x}) = f_k^* - \sum_{j=1}^{n}\widetilde{c}_{kj}x_j$, $\widetilde{B}_k(\mathbf{x})$ is normally distributed, and let $E\left(\widetilde{B}_k(\mathbf{x})\right)$ and $\text{Var}\left(\widetilde{B}_k(\mathbf{x})\right)$ be, respectively, the mean and the variance. Thus, we have:

$$\text{Prob}(\widetilde{B}_k(\mathbf{x}) \le y) \ge 1-\alpha_k,\ k=1,\ldots,K, \tag{22}$$

$$\text{Prob}\left(\frac{\widetilde{B}_k(\mathbf{x}) - E\left(\widetilde{B}_k(\mathbf{x})\right)}{\sqrt{\text{Var}\left(\widetilde{B}_k(\mathbf{x})\right)}} \le \frac{y - E\left(\widetilde{B}_k(\mathbf{x})\right)}{\sqrt{\text{Var}\left(\widetilde{B}_k(\mathbf{x})\right)}}\right) \ge 1-\alpha_k,\ k=1,\ldots,K, \tag{23}$$

where $\Phi(z)=P(N(0,1)\le z)$ represents the cumulative distribution function of a standard normal distribution.

$$\frac{y - E\left(\widetilde{B}_k(\mathbf{x})\right)}{\sqrt{\text{Var}\left(\widetilde{B}_k(\mathbf{x})\right)}} \ge \Phi^{-1}(1-\alpha_k),\ k=1,\ldots,K, \tag{24}$$

$$y \ge E\left(\widetilde{B}_k(\mathbf{x})\right) + \Phi^{-1}(1-\alpha_k)\sqrt{\text{Var}\left(\widetilde{B}_k(\mathbf{x})\right)},\ k=1,\ldots,K, \tag{25}$$

$$y \ge E\left(f_k^* - \sum_{j=1}^{n}\widetilde{c}_{kj}x_j\right) + \Phi^{-1}(1-\alpha_k)\sqrt{\text{Var}\left(f_k^* - \sum_{j=1}^{n}\widetilde{c}_{kj}x_j\right)},\ k=1,\ldots,K. \tag{26}$$

Therefore with regard to Eqs. 17 and 26 and by considering preference weights of the random objectives and constraints, the CCGC (min–max) model, which is equivalent to program 2, can be stated as:

$$\begin{aligned} &\min\ y,\\ &\text{subject to}\\ &y \ge w_k \times \left(E\left(f_k^* - \sum_{j=1}^{n}\widetilde{c}_{kj}x_j\right) + \Phi^{-1}(1-\alpha_k)\sqrt{\text{Var}\left(f_k^* - \sum_{j=1}^{n}\widetilde{c}_{kj}x_j\right)}\right),\ k=1,\ldots,K,\\ &y \ge w_i \times \left(E\left(\widetilde{b}_i - \sum_{j=1}^{n}\widetilde{a}_{ij}x_j\right) - \Phi^{-1}(1-\alpha_i)\sqrt{\text{Var}\left(\sum_{j=1}^{n}\widetilde{a}_{ij}x_j - \widetilde{b}_i\right)}\right),\ i=1,\ldots,m,\\ &\sum_{j=1}^{n}x_j = X,\\ &\mathbf{x}\in S,\\ &y\ge 0, \end{aligned} \tag{27}$$

where $\sum_{k=1}^{K} w_k + \sum_{i=1}^{m} w_i = 1$ (w_k, $w_i>0$, for $k=1,\ldots,K$ and $i=1,\ldots,m$).

In the next section, the model proposed will be illustrated through a multi-objective problem of determining optimal number of manpower for a job-shop system. Then, the results obtained from solving the multi-objective problem by the model proposed and the CCCP (min–max) model are compared and evaluated.

6 Numerical example

The purpose of this numerical case is to show how the proposed model can be applied to solve manpower allocation problem under uncertainty. It is assumed that there is a job-shop system where two independent workshops with several work units are active. The aim is to determine the optimal number of manpower in the system by considering some random parameters. The other assumptions for the problem are as below.

6.1 Assumptions

1. The final product related to jth unit (for $j=1,\ldots,n$) is computed by Eq. 28:

$$(\text{Output})_j = (\text{Input})_j - (\text{Failure})_j, \tag{28}$$

2. The failed parts are not reworked but they are assumed to be scraped.
3. The number of daily failed parts of jth unit is random variable and normally distributed with known mean and variance.
4. In this system, there is no reworking process.
5. Based on kind of activities in each work unit, several workstations are active.
6. Here, a workstation is defined as a place that is occupied by a set or work seat, equipments, needed tools, and one worker.
7. The holding cost of each unit of final product in warehouse is insignificant.
8. There is not any constraint for the holding space of final products in warehouse.
9. In optimal solution, if optimal number of manpower related to jth unit is a decimal value of less than 0.5, then it rounds down to the nearest integer value and if it is above 0.5, it rounds up.
10. The quantity of daily products has a direct relation with the number of manpower in the system.
11. The assembling station is completely automatic.
12. Kind of products produced in the two workshops are different.

6.2 Determining the optimal number of manpower in workshop 1

In this manufacturing system, workshop 1 is managed based on the job-shop system. In this workshop, five work units have been independently located in parallel (Fig. 1). The objectives and constraints considered for solving the problem of determining optimal number of manpower in the manufacturing system are as below.

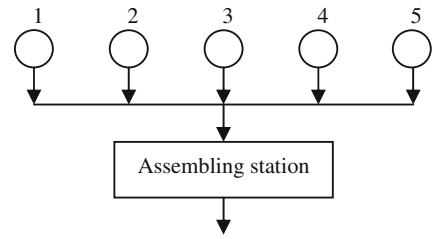

Fig. 1 Workshop 1 with five independent work units is managed based on job-shop system

6.2.1 Objectives

1. The stochastic objective of final product (output):

$$\widetilde{f}_1(\mathbf{x}) = \sum_{j=1}^{5} \widetilde{o}_j x_j$$

where

$\widetilde{o}_j$ random variable of output quantity per worker in jth unit

x_j decision variable of number of the manpower in jth unit

In order to utilize all the capacity of the manufacturing system and to decrease the times of setup and shortage of products as well, the aim of maximizing the daily final product in the system is desired.

2. The daily wage cost objective of workers:

$$f_2(\mathbf{x}) = \sum_{j=1}^{5} w_j x_j$$

where

w_j daily wage/worker employed in work unit jth

Based on degree of being professional and the skill needed in jth unit, the daily wage/worker employed in different units is diversity. The aim is to minimize this objective.

The capability of the proposed model speaks of the fact that there is no limitation as regards the value of the coefficients of objective function. Nonetheless, the present study assumes that the coefficients of wage cost objective function are specified and final according to the available information, and by virtue of these data, an objective function can be prepared for optimally determining new manpower number.

3. The stochastic objective of total unallowable idle times of workers:

$$\widetilde{f}_3(\mathbf{x}) = \sum_{j=1}^{5} \widetilde{t}_j x_j$$

where

$\widetilde{t}_j$ random variable of unallowable idle time/worker employed in jth unit

The aim is to minimize total idle times of daily unallowable in the manufacturing system.

6.2.2 Constraints

1. With regard to constraint of needed manpower location space, the minimum number of manpower located in jth unit is equal to number of existent manpower in this unit. Also, the maximum manpower number of jth unit is equal to nine workers. Table 1 presents the number of existent workers employed ($x_{j(\text{existent})}$) in the different units of workshop 1.
2. The total employed workers in workshop 1 are equal to 30 workers, where $\sum_{j=1}^{5} x_j = 30$.

Table 1 The number of existent workers employed in the work units of workshop 1

Work unit (j)	1	2	3	4	5
$x_{j(\text{existent})}$	3	3	4	2	3

At present, from among 30 workers in workshop 1, 15 workers have been employed as existent manpower and 15 workers will be employed as new manpower in the future. The aim is to determine optimal manpower in each work unit by considering optimal achievement to the objectives and the constraints defined in the system.

The main tri-objective stochastic problem of determining the optimal number of manpower in located work units in workshop 1 is as follows:

$$\begin{aligned}
&\max\ \widetilde{f}_1(\mathbf{x}) = \textstyle\sum_{j=1}^{5} \widetilde{o}_j x_j,\\
&\min\ f_2(\mathbf{x}) = \textstyle\sum_{j=1}^{5} w_j x_j,\\
&\min\ \widetilde{f}_3(\mathbf{x}) = \textstyle\sum_{j=1}^{5} \widetilde{t}_j x_j,\\
&\text{subject to}\\
&x_1 + x_2 + x_3 + x_4 + x_5 = 30,\\
&3 \le x_1 \le 9,\\
&3 \le x_2 \le 9,\\
&4 \le x_3 \le 9,\\
&2 \le x_4 \le 9,\\
&3 \le x_5 \le 9.
\end{aligned} \tag{29}$$

The data used for the tri-objective stochastic problem are based on daily information related to work-year of the manufacturing system, and it has been presented in Table 2. Table 2 shows the best values of the coefficients of the objectives for work units of workshop 1. Other needed information of the problem there is in program 30. Also the utopia values of all objectives of program 29 with regard to solution space of this program have been presented in Table 3.

Based on the CCGC (min–max) model, the tri-objective stochastic problem can be transformed to a deterministic equivalent model as follows:

$$\begin{aligned}
&\min\ \ y,\\
&\text{subject to}\\
&y \ge w_1 \times \Big((362 - (8x_1 + 5x_2 + 12x_3 + 6x_4 + 10x_5)) + \Phi^{-1}(1-\alpha_1)\\
&\qquad \times \sqrt{0.000125x_1^2 + 0.000324x_2^2 + 0.000469x_3^2 + 0.000521x_4^2 + 0.000324x_5^2} \Big),\\
&y \ge w_2 \times (20x_1 + 15x_2 + 17x_3 + 12x_4 + 18x_5 - 459),\\
&y \ge w_3 \times \Big((40x_1 + 60x_2 + 35x_3 + 50x_4 + 45x_5 - 1{,}067) + \Phi^{-1}(1-\alpha_3)\\
&\qquad \times \sqrt{0.000162x_1^2 + 0.00021x_2^2 + 0.000135x_3^2 + 0.000222x_4^2 + 0.000198x_5^2} \Big),\\
&x_1 + x_2 + x_3 + x_4 + x_5 = 30,\\
&3 \le x_1 \le 9,\\
&3 \le x_2 \le 9,\\
&4 \le x_3 \le 9,\\
&2 \le x_4 \le 9,\\
&3 \le x_5 \le 9,\\
&y \ge 0.
\end{aligned} \tag{30}$$

Table 2 Information related to the work units of workshop 1

Work unit (j)	1	2	3	4	5
Mean of daily output quantity/worker	8	5	12	6	10
Maximum of daily output quantity/worker	10	8	15	8	13
Daily wage cost/worker	20	15	17	12	18
Mean of daily unallowable idle time/worker	40	60	35	50	45
Minimum of daily unallowable idle time/worker	30	48	32	46	39

Also, the stochastic problem, based on the CCCP (min–max) model, can be transformed to a deterministic equivalent model as program 31:

$$
\begin{aligned}
&\min \quad y,\\
&\text{subject to}\\
&\Big((362-(8x_1+5x_2+12x_3+6x_4+10x_5))+\Phi^{-1}(1-\alpha_1)\\
&\qquad\times\sqrt{0.000125x_1^2+0.000324x_2^2+0.000469x_3^2+0.000521x_4^2+0.000324x_5^2}\Big)\\
&\qquad-\epsilon_1+p_1=0,\\
&20x_1+15x_2+17x_3+12x_4+18x_5-n_2=459,\\
&\Big((40x_1+60x_2+35x_3+50x_4+45x_5-1{,}067)+\Phi^{-1}(1-\alpha_3)\\
&\qquad\times\sqrt{0.000162x_1^2+0.00021x_2^2+0.000135x_3^2+0.000222x_4^2+0.000198x_5^2}\Big)\\
&\qquad-\epsilon_3+p_3=0,\\
&y\geq w_1\times(\epsilon_1+p_1),\\
&y\geq w_2\times(n_2),\\
&y\geq w_3\times(\epsilon_3+p_3),\\
&x_1+x_2+x_3+x_4+x_5=30,\\
&3\leq x_1\leq 9,\\
&3\leq x_2\leq 9,\\
&4\leq x_3\leq 9,\\
&2\leq x_4\leq 9,\\
&3\leq x_5\leq 9,\\
&y\geq 0;\ \epsilon_1,\epsilon_3\geq 0;\ p_1,p_3\geq 0;\ n_2\geq 0.
\end{aligned}
\tag{31}
$$

In programs 30 and 31, w_1=w_2=0.4, w_3=0.2, and α_1=α_3=0.05 are considered. Programs 30 and 31 have been solved and the obtained results from solving them are shown in Table 4.

Table 3 The utopia values of objectives in solution space of the tri-objective stochastic problem

Objective	Utopia value
Output	362
Wage cost	459
Unallowable idle time	1,067

Table 4 presents the obtained results of the stochastic problem by using the CCGC (min–max) and CCCP (min–max) models, which are both feasible and optimal. In Table 4, vector of values x_j (for j=1,…, 5) using the CCGC (min–max) and CCCP (min–max) models is as (9, 3, 9, 2, 7). The strategy of supplying human resources, which is related to solving the above tri-objective problem, is as follows:

– Work unit 1: Six workers will be employed as new manpower.

Table 4 The results of solving the problem of optimal number of manpower in work units of workshop 1

	Existent system	CCGC (min–max)	CCCP (min–max)
x_1	3	9	9
x_2	3	3	3
x_3	4	9	9
x_4	2	2	2
x_5	3	7	7
Output	129	277	277
Wage cost	251	528	528
Unallowable idle time	675	1,270	1,270
Number of variables		6	11
Number of constraints		9	12
Number of iterations		15	19

- Work unit 2: The system activity continues with these existent three workers.
- Work unit 3: Five workers will be employed as new manpower.
- Work unit 4: The system activity continues with these existent two workers.
- Work unit 5: Four workers will be employed as new manpower.

The vector of objectives' optimal values that have obtained by programs 30 and 31 is such as (277, 528, 1,270), so that nearly all objectives, the CCGC (min–max) and CCCP (min–max) models, wholly present the same results. This can demonstrate the similarity of two models while the CCGC (min–max) has been formulated with fewer variables and constraints. In Table 4, the results obtained from existent situation of the system for the objectives of problem are equal to (129, 251, 675). Also the necessary iteration number for achieving optimal solution of the problem is equal to 15 and 19 iterations that, respectively, have obtained from solving programs 30 and 31. Hence, it can be said that with the problem dimensions increased (increasing the number of variables and constraints of the tri-objective problem), CCGC (min–max) model is able to achieve the optimal solution with a short solution time and a less complicacy.

In this problem, the stochastic objectives of final output and the unallowable idle time and the deterministic objective of wage cost were introduced. The aim of tri-objective problem is to extend the manufacturing system and to increase accessible quota in competing market. On the other side, with manpower being decreased (or increased), the quantity of final output of system decreases (or increases), the attracting and employing system will decrease (or increase), and the total unallowable idle times of system manpower decreases (or increases). About the stochastic problem, $w_1=w_2>w_3$ has been considered, which the optimal values of objectives output and wage cost under two situations, that is, before and after optimizing, have been presented in Table 5.

Table 5 Optimal values of the objectives of output and wage cost under two situations

Objective	Before optimizing	After optimizing	Amount of variations
Output	129	277	148
Wage cost	251	528	277

For this, we can analyze the obtained results by solving the tri-objective problem under these two situations: The situation of before optimizing states the existent situation of workshop 1 by considering minimum active worker in different work units and the situation of after optimizing states the results of CCCP (min–max) and CCGC (min–max) models in Table 4. Also, the value of variations of obtained results indicates the difference between results of these two situations. Figure 2 shows the optimal values of objectives output and wage cost under two situations of before and after optimizing programs 30 and 31.

According to Fig. 2, under pre-optimizing conditions, the system has minimum paying wage and minimum output, which results in minimum income obtained from selling products. But under situation of after optimizing, the system has maximum paying wage and maximum output, so it will simultaneously increase the system cost and income.

The obtained results, under two pre and post-optimization conditions, represent the kind of trade-off between objectives output and wage cost, which will cause planners to easily make decisions on two employment and unemployment situations of manpower under uncertainty. As Table 5 and Fig. 2 show, the variations in objectives of output and wage cost are linear.

6.3 Determining optimal number of manpower employed in workshops 1 and 2

Now in order to survey performance of the proposed model, we should optimize the summation of manpower employed in two workshops 1 and 2. The data related to workshop 1 have been presented in Tables 1 and 2. In workshop 2, there are three work units, whose units are

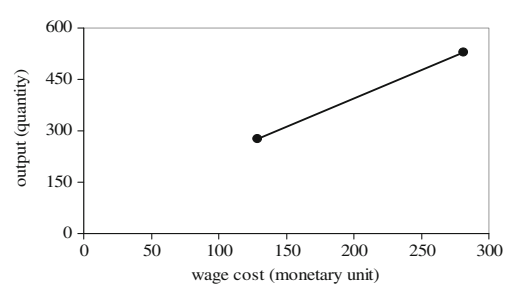

Fig. 2 Variations of the objectives of output and wage cost based on the existent system and after optimizing the stochastic problem by CCGC (min–max) model

Table 6 The data associated with the work units of workshop 2

Work unit (j)	6	7	8
$E(\widetilde{o}_j)$	10	9	11
$\mathrm{Var}(\widetilde{o}_j)$	0.000436	0.000339	0.000215
$\max(\widetilde{o}_j)$	13	12	16
w_j	18	15	17
$E(\widetilde{t}_j)$	30	40	40
$\mathrm{Var}(\widetilde{t}_j)$	0.000231	0.000225	0.000301
$\min(\widetilde{t}_j)$	26	32	36

considered by numbers 6, 7, and 8, respectively, for simplicity of computations. Information associated with workshop 2 in the same time interval is shown in Table 6. Table 6 shows the best values of the coefficients of the objectives for work units of workshop 2 where $3 \le x_6, x_7 \le 9$, $4 \le x_8 \le 9$, $w_1 = w_2 = 0.4$, $w_3 = 0.2$, and $\alpha_1 = \alpha_3 = 0.05$ are considered. Constraint of manpower employed in the manufacturing system is equal to $\sum_{j=1}^{8} x_j = 40$.

Again, the tri-objective stochastic problem with the new variables and constraints is solved, and the obtained results are presented in Table 7. With regard to Table 7, the results of optimizing the tri-objective stochastic problem by each of CCGC (min–max) and CCCP (min–max) models are completely equivalent and $f_1^* = 395$, $f_2^* = 678$, and $f_3^* = 1585$.

As we saw, with work units increased from 5 to 8, the optimal value of objectives will be increased. In the other words, with increase of work units that leads to increase daily output, the paid wage cost to workers and their total unallowable idle time will increase, too. In Table 7, with work units increased from 5 to 8, the number of variables, constraints, and iterations in CCGC (min–max) model compared to CCCP (min–max) model for solving the tri-objective problem decreases (see the bolded values in Table 7), and this shows that to achieve the same results, CCGC (min–max) model in modeling the tri-objective problem based on five and eight work units has a less complicacy and solving that problem has a less number of iteration.

6.4 Experimental analysis

To further analyze the results and examine the capabilities of the proposed model for solving different sized problems, some assumptions of the programs 30 and 31 are changed, and results of solving the tri-objective stochastic problem are examined based on the new assumptions. At this stage, executive steps of both programs 30 and 31 are as follows:

1. Coefficients of the first, second, and third objectives are uniformly distributed in the range of [6, 15], [10, 20], and [30, 60], respectively.
2. Threshold values are uniformly distributed in the range of [0.01, 0.1].
3. Preference weights for the objectives are uniformly distributed in the range of [0.01, 0.99].
4. Total number of workers is uniformly distributed in the range of [30, 335].
5. Lower and upper bound of workers in each work unit is uniformly distributed in the range of [2, 4] and [8, 10], respectively.
6. The experiment begins with five work units and then increases systematically by the increment of 5 to the maximum number of 100 work units. Therefore, the experiment includes 20 random problems. Each problem is solved five times to account for the random changes of the parameter values.

The results of the experiment for the tri-objective manpower allocation problem are presented in Table 8. Figure 3 shows the changes in average number of iterations required for solving 20 tri-objective problems.

Table 8 shows that results obtained by CCCP and CCGC methods are similar for each problem but numbers of iterations required for solution are different. According to Fig. 3, the average number of iterations required for solving the problem by CCGC method is less than CCCP method. As the problem input size increases, the average number of iterations required for solution remains almost unchanged for CCGC method. The results show that considering all random conditions of tri-objective stochastic problem, CCGC method could present similar results with less number of iterations compared to CCCP method, which could be of significant advantage for solving the large-scale problems.

The optimality of the proposed model is considered from two views:

1. Quality of solution: As it was stated, the results obtained by the proposed model are thoroughly similar to the results of CCCP. Therefore, considering the authenticity of the modeling of CCCP model and the results of it, the

Table 7 The obtained results from solving the problem of determining optimal number of manpower for workshops 1 and 2

Model	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	Number of variables	Number of constraints	Number of iterations
CCCP (min–max)	3	3	9	2	4≈4.17	7≈6.83	3	9	14	15	25
CCGC (min–max)	3	3	9	2	4≈4.17	7≈6.83	3	9	9	12	15

Table 8 Results of the experiment with respect to the number of iterations

Problem no.	Minimum		Average		Maximum	
	CCGC	CCCP	CCGC	CCCP	CCGC	CCCP
1	10	21	10	22.8	10	29
2	9	10	9	18.2	9	31
3	9	12	10.6	22.6	12	33
4	8	36	9.6	36.2	10	37
5	8	9	8.4	9.8	10	13
6	8	10	8.2	16.8	9	26
7	8	11	8	11	8	11
8	3	3	3.8	3	4	3
9	8	12	8	12	8	12
10	18	22	18.2	23.2	19	24
11	15	25	17.6	18.4	19	35
12	15	30	18.2	31.8	20	36
13	19	27	19.6	52.2	20	125
14	14	32	19.6	51.2	29	122
15	14	28	19.2	30.6	21	37
16	16	28	18.4	29	21	34
17	16	29	18	30.2	24	33
18	14	27	18	29.4	20	34
19	13	23	15.8	23.6	18	25
20	16	22	16.4	23	18	24

quality of the results of the proposed model can be confirmed as well.

2. Method for obtaining solution: Section 3 presented the modeling method of GC. GC model, exerting some changes in CP model, can be formulated based on lower variables and constraints. As mentioned earlier, the number of variables and constraints of GC model is lower than that of CP model, and such a decrease can be effective in the method of attaining results. As mentioned in the paper, the results of the proposed model are thoroughly similar to that of CCCP save for decreased iterations of solution in the proposed model. This can be seen in the results obtained for 20 problems with different dimensions, which is shown in Fig. 3.

This capability of the proposed model can be considered for solving large-scale problems in shorter times (lower iterations).

7 Conclusion

This paper introduced a model called CCGC (min–max) that was a deterministic equivalent for solving multi-objective stochastic problems through combining the CCP approach and the GC (min–max) model. The proposed model was illustrated by a tri-objective stochastic problem for determining optimal number of manpower in a manufacturing system of job-shop with two workshops and several work units. The random variables of stochastic problem were the quantity of output produced and the lost unallowable idle time by workers. The aim of this system is to develop and to influence the competing market.

Initially, the stochastic problem based on information of workshop 1 was formulated, and the problem was solved by CCGC (min–max) and CCCP (min–max) models. The results obtained from solving the problem by these models showed feasibility, optimality, and similarity of the solutions. We saw that CCGC (min–max) model with less variables and constraints has less iteration for finding the optimal solution. This could be because of decreasing the number of variables and constraints in modeling of CCGC (min–max) model.

Again, the tri-objective stochastic problem was formulated for both workshops 1 and 2. The results obtained from solving the problem by CCGC (min–max) model showed that increasing the number of variables and constraints had not much effect on the number of necessary iterations for finding the optimal solution and this model decreases complicacy in modeling the problem. For more confidence about performance efficiency of the proposed model, 20 different problems were solved by considering new assumptions used

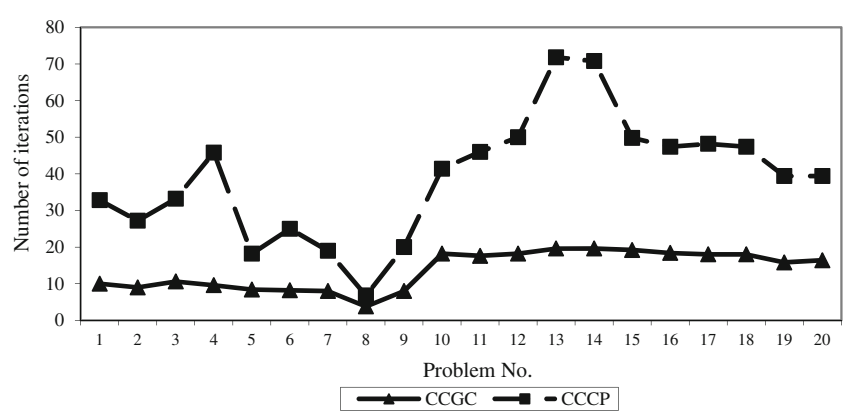

Fig. 3 Variations of average number of iterations

in CCGC (min–max) and CCCP (min–max) models, and because of the similarity among the obtained results in different assumptions, mean number of iterations for solving by CCGC (min–max) were fewer than CCCP (min–max).

Decreasing solution time and increasing accuracy of modeling for stochastic large-scale problems with large scales have always been important for planners. Hence, decreasing number of variables and constraints in CCGC (min–max) model can be a suitable approach for achieving these objectives. The proposed model can be easily adapted to other areas of operation research and management sciences where optimizing a multiple objective problem under uncertainty are of high interest, such as financial planning, portfolio optimization, inventory management, supply chain management, and resource allocation.

Acknowledgments The authors are grateful for the valuable comments received from the unknown reviewers. The authors believe that the changes in the original manuscript have made it significantly improved.

References

1. Huang DK, Chiu HN, Yeh RH, Chang JH (2009) A fuzzy multi-criteria decision making approach for solving a bi-objective personnel assignment problem. Comput Ind Eng 56:1–10
2. Caballero R, Galache T, Gómez T, Molina J, Torrico A (2004) Budgetary allocations and efficiency in the human resources policy of a university following multiple criteria. Econ Educ Rev 23:67–74
3. Kwak NK, Lee C (1997) A linear goal programming model for human resource allocation in a health-care organization. J Med Syst 21:129–140
4. Summers EL (1972) The audit staff assignment problem: a linear programming analysis. Account Rev 47:443–453
5. Killough LN, Souders TL (1973) A goal programming model for public accounting firms. Account Rev XLVIII(2):268–279
6. Bailey AD, Boe WJ, Schnack T (1974) The audit staff assignment problem: a comment. Account Rev LIX(3):572–574
7. Welling P (1977) A goal programming model for human resource accounting in a CPA firm. Account Organ Soc 2:307–316
8. Balachandran KR, Steuer RE (1982) An interactive model for the CPA firm audit staff planning problem with multiple objectives. Account Rev LVII(1):125–140
9. Andersson J (2000) A survey of multiobjective optimization in engineering design. Optimization (LiTH-IKP-R-1097), 34.Citeseer. Retrieved from: http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.8.5638&rep=rep1&type=pdf
10. Kwak W, Shi Y, Jung K (2003) Human resource allocation in a CPA firm: a fuzzy set approach. Rev Quant Finan Acc 20:277–290
11. Gardner JC, Huefner RJ, Lotfi V (1990) A multi period audit staff planning model using multiple objectives: development and evaluation. Decis Sci 21:154–170
12. Chebat J-C, Filiatrault P, Katz A, Tal SM (1994) Strategic auditing of human and financial resource allocation in marketing an empirical study using data envelopment analysis. J Bus Res 31:197–208
13. Pan Q-K, Suganthan PN, Chua TJ, Cai TX (2010) Solving manpower scheduling problem in manufacturing using mixed-integer programming with a two-stage heuristic algorithm. Int J Adv Manuf Technol 46:1229–1237
14. Liang GS, Wang MJ (1992) Personnel placement in a fuzzy environment. Comput Oper Res 19:107–121
15. Zadeh LA (1965) Fuzzy sets. Inf Control 8:338–353
16. Toroslu IH (2003) Personnel assignment problem with hierarchical ordering constraints. Comput Ind Eng 45:493–510
17. Liang GS, Wang MJ (1994) Personnel selection using fuzzy MCDM algorithm. Eur J Oper Res 78:22–33
18. Yaakob SB, Kawata S (1999) Workers' placement in an industrial environment. Fuzzy Set Syst 106:289–297
19. Karsak EE (2000) A fuzzy multiple objective programming approach for personnel selection. IEEE International Conference on Systems, Man, and Cybernetics, vol 3, pp 2007–2012
20. Abdelaziz FB, Aouni B, Fayedh RE (2007) Multi-objective stochastic programming for portfolio selection. Eur J Oper Res 177:1811–1823
21. Sen S (2001) In: Gass S, Harris C (eds) Stochastic programming: computational issues and challenges, encyclopedia of OR/MS. Kluwer Academic, Dordrecht, pp 784–789
22. Murphy FH, Sen S, Soyster AL (1982) Electric utility capacity expansion planning with uncertain load forecasts. AIIE Trans 14:52–59
23. Carino DR, Kent T, Meyers DH, Stacy C, Sylvanus M, Turner AL, Watanabe K, Ziemba WT (1994) The Russell–Yasuda Kasai model: an asset/liability model for a Japanese insurance company using multistage stochastic programming. Interfaces 24(1):29–49
24. Sen S, Doverspike RD, Cosares S (1994) Network planning with random demand. Telecommun Syst 3:11–30
25. Ip WH, Fung R, Keung KW (1999) An investigation of stochastic analysis of flexible manufacturing systems simulation. Int J Adv Manuf Technol 15:244–250
26. Fisher ML, Hammond J, Obermeyer W, Raman A (1997) Configuring a supply chain to reduce the cost of demand uncertainty. Prod Oper Manag 6:211–225
27. Lim SJ, Jeong SJ, Kim KS, Park MW (2006) Hybrid approach to distribution planning reflecting a stochastic supply chain. Int J Adv Manuf Technol 28:618–625
28. Sheikh Sajadieh M, AkbariJokar MR (2009) An integrated vendor–buyer cooperative model under stochastic supply lead-time. Int J Adv Manuf Technol 41:1043–1050
29. Mirzapour Al-e-Hashem SMJ, Aryanezhad MB, Sadjadi SJ (2011) An efficient algorithm to solve a multi-objective robust aggregate production planning in an uncertain environment. Int J Adv Manuf Technol. doi:10.1007/s00170-011-3396-1
30. Kall P, Wallace SW (1994) Stochastic programming. Wiley, England
31. Birge JR, Louveaux F (1997) Introduction to stochastic programming. Springer, New York
32. Prékopa A (1995) Stochastic programming. Kluwer Academic, London
33. Birge JR (1997) Stochastic programming computation and applications. INFORMS J Comput 9:111–133
34. Sen S, Higle JL (1999) An introductory tutorial on stochastic linear programming models. Interfaces 29:33–61
35. Charnes A, Cooper WW (1959) Chance constrained programming. Manag Sci 6:73–79
36. Charnes A, Cooper WW (1963) Deterministic equivalents for optimizing and satisfying under chance constraints. Oper Res 11:18–39
37. Ahmadizar F, Ghazanfari M, FatemiGhomi SMT (2009) Application of chance-constrained programming for stochastic group shop scheduling problem. Int J Adv Manuf Technol 42:321–334
38. Lawrence SR, Sewell EC (1997) Heuristic, optimal, static, and dynamic schedules when processing times are uncertain. J Oper Manag 15:71–82
39. Singer M (2000) Forecasting policies for scheduling a stochastic due date job shop. Int J Prod Res 38:3623–3637

40. Yoshitomi Y (2002) A genetic algorithm approach to solving stochastic job-shop scheduling problems. Int Trans Oper Res 9:479–495
41. Yoshitomi Y, Yamaguchi R (2003) A genetic algorithm and the Monte Carlo method for stochastic job-shop scheduling. Int Trans Oper Res 10:577–596
42. Tavakkoli-Moghaddam R, Jolai F, Vaziri F, Ahmed PK, Azaron A (2005) A hybrid method for solving stochastic job shop scheduling problems. Appl Math Comput 170:185–206
43. Luh PB, Chen D, Thakur LS (1999) An effective approach for job-shop scheduling with uncertain processing requirements. IEEE Trans Robot Autom 15:328–339
44. Neumann K, Schneider WG (1999) Heuristic algorithms for job shop scheduling problems with stochastic precedence constraints. Ann Oper Res 92:45–63
45. Alcaide D, Rodriguez-Gonzalez A, Sicilia J (2006) A heuristic approach to minimize expected make span in open shops subject to stochastic processing times and failures. Int J Flex Manuf Syst 17:201–226
46. Athan TW, Papalambros PY (1996) A note on weighted criteria methods for compromise solutions in multi-objective optimization. Eng Optim 27:155–176
47. Zeleny M (1973) Compromise programming. In: Cochrane JL, Zeleny M (eds) Multiple criteria decision making. University of South Carolina Press, Columbia, pp 262–301
48. Zeleny M (1974) A concept of compromise solutions and the method of the displaced ideal. Comput Oper Res 1:479–496
49. Yu PL (1973) A class of solutions for group decision problems. Manag Sci 19:936–946
50. Zeleny M (1976) Multiple criteria decision making. Springer, Berlin
51. Zeleny M (1982) Multiple criteria decision making. McGraw-Hill, New York
52. Osyczka A (1984) Multicriterion optimization in engineering with FORTRAN programs. Wiley, New York
53. Eschenauer H, Koski J, Osyczka A (1990) Multicriteria design optimization procedures and applications. Springer, Berlin
54. Miettinen K (1999) Nonlinear multi-objective optimization. Kluwer Academic, Boston
55. Marler RT (2005) A study of multi-objective optimization methods for engineering applications. A thesis submitted in partial fulfillment of the requirements for the Doctor of Philosophy degree in Mechanical Engineering in the Graduate College of the University of Iowa